

# ImageSNN Evaluation Report

Generated: 12 August 2026 at 11:30  
Images Tested: 700  
Model Version: 1.0.0  
Generation Manifest: generation\_manifest\_20260715\_095824.json

## Evaluation Summary

---

| Metric    | Value  |
|-----------|--------|
| Accuracy  | 0.7729 |
| Precision | 0.8848 |
| Recall    | 0.8450 |
| F1 Score  | 0.8645 |
| AUC       | 0.6344 |
| MSE       | 0.2200 |

## Key Findings

---

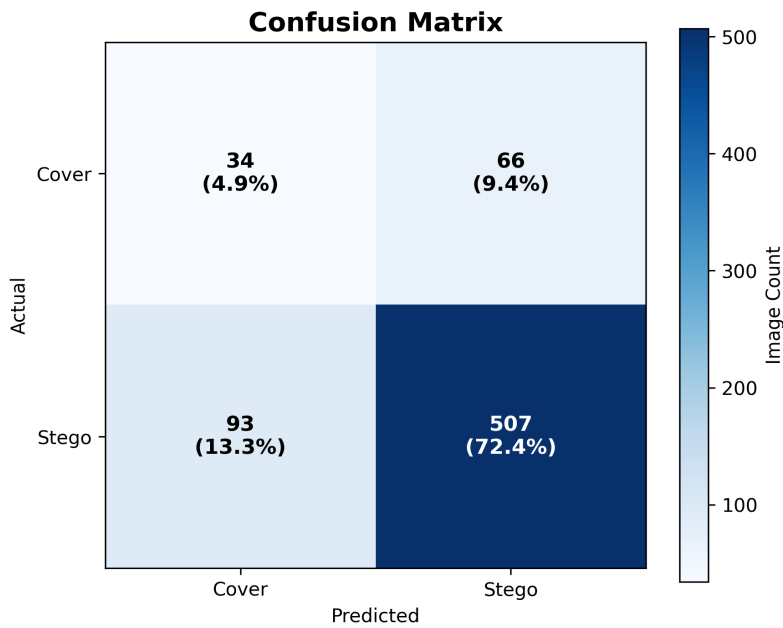
- Model accuracy achieved 77.29%.
- Recall reached 84.50%, indicating successful detection of most stego images.
- Error rate was 22.71% across analysed images.
- Mean inference latency was 258.74 ms with a throughput of 3.86 images/s.

## Validation Summary

---

Validation reporting will be added in a future release.

# Confusion Matrix



## Summary

True Positives: 507

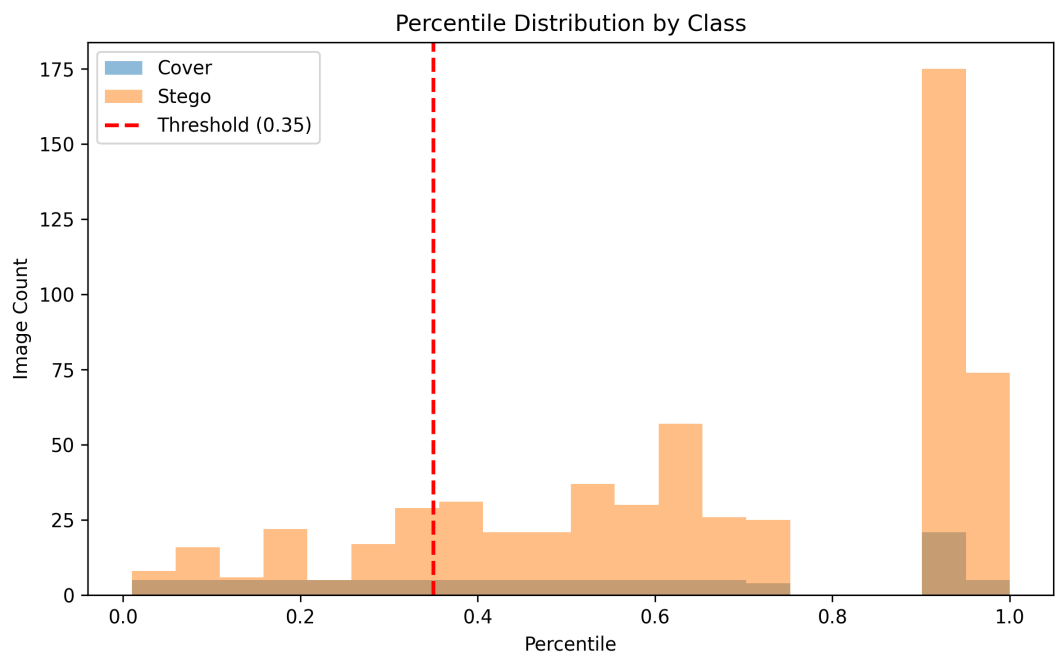
True Negatives: 34

False Positives: 66

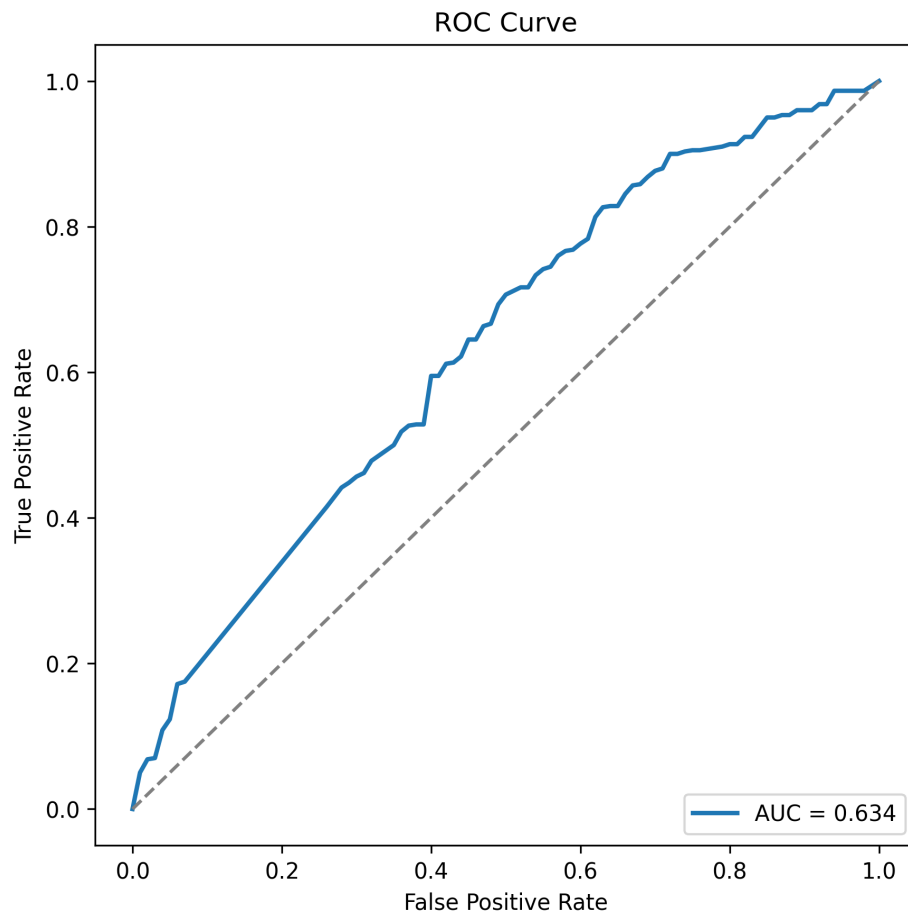
False Negatives: 93

The model correctly classified 541 images and misclassified 159 images across the evaluation dataset.

# Probability Distribution

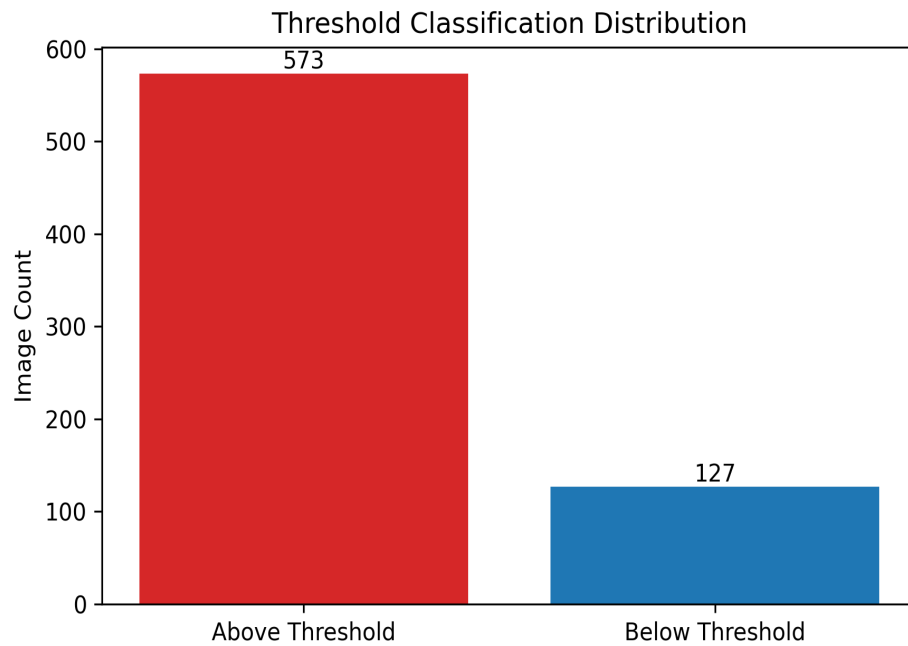


## ROC Curve



## Threshold Distribution

---



### Summary

Above Threshold: 573

Below Threshold: 127

Threshold Proportion: 0.82%

0.82% of predictions exceeded the configured decision threshold.

## Classification Outcomes

---

Mean Difference: 0.1389  
False Positives: 66  
False Negatives: 93  
Above Threshold: 573  
Below Threshold: 127  
Threshold Proportion: 81.86%

## Error Analysis Summary

---

### Classification Results

| Metric          | Value |
|-----------------|-------|
| Pairs Analysed  | 600   |
| True Positives  | 507   |
| False Positives | 66    |
| True Negatives  | 34    |
| False Negatives | 93    |
| Total Errors    | 159   |

### Performance Metrics

| Metric            | Value  |
|-------------------|--------|
| TPR (Recall)      | 84.50% |
| TNR (Specificity) | 34.00% |
| FPR               | 66.00% |
| FNR               | 15.50% |
| Error Rate        | 22.71% |

## Dataset Quality

---

| Folder                 | Avg PSNR | Avg SSIM |
|------------------------|----------|----------|
| <b>Stego Encrypted</b> |          |          |
| Large                  | 56.2087  | 0.998619 |
| Medium                 | 61.2381  | 0.999562 |
| Small                  | 66.5664  | 0.999871 |
| <b>Stego Plain</b>     |          |          |
| Large                  | 56.1952  | 0.998616 |
| Medium                 | 61.2893  | 0.999568 |
| Small                  | 66.5479  | 0.999870 |

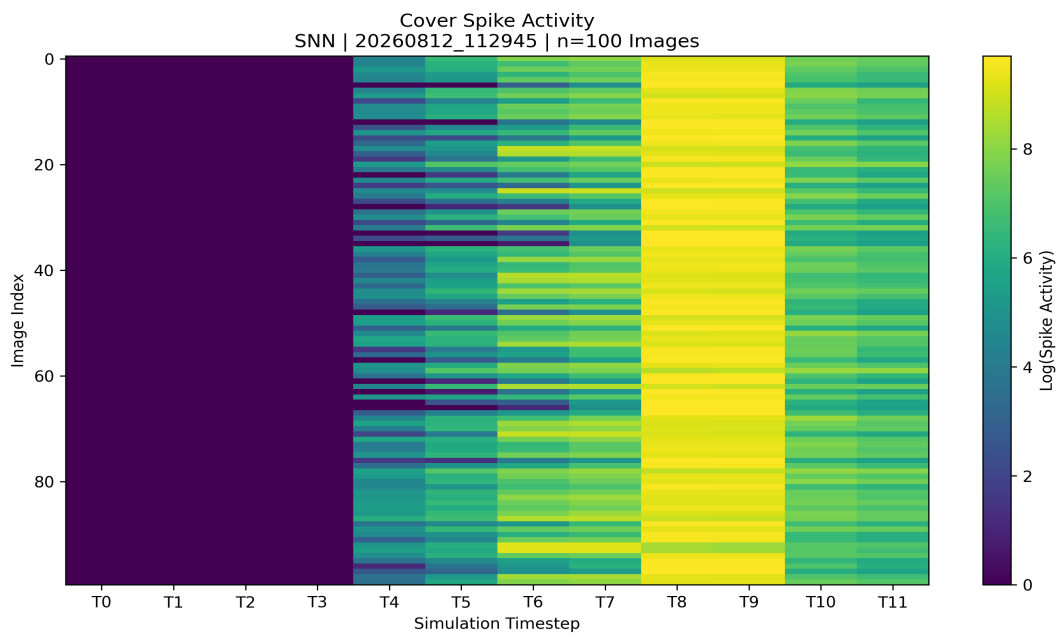
*PSNR and SSIM metrics were used to assess visual similarity between cover and stego images. Higher PSNR and SSIM values indicate lower visual distortion and greater similarity to the original cover images.*

*Average PSNR and SSIM values are presented for reporting purposes. Detailed quality statistics, including minimum and maximum measurements, are provided within the accompanying workbook export.*

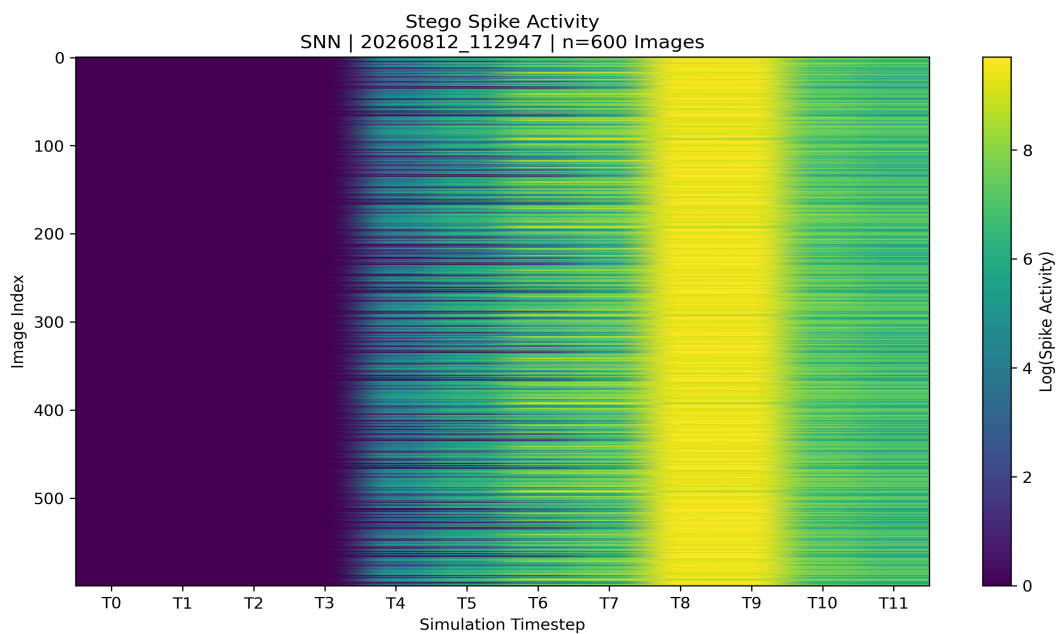
## SNN Decision Analysis

Spike activation visualisations were generated to examine temporal firing behaviour within the hidden LIF layer of the ImageSNN architecture.

### Cover Spike Activation Map



### Stego Spike Activation Map





## Key Findings

Mean Spike Difference: -0.0700

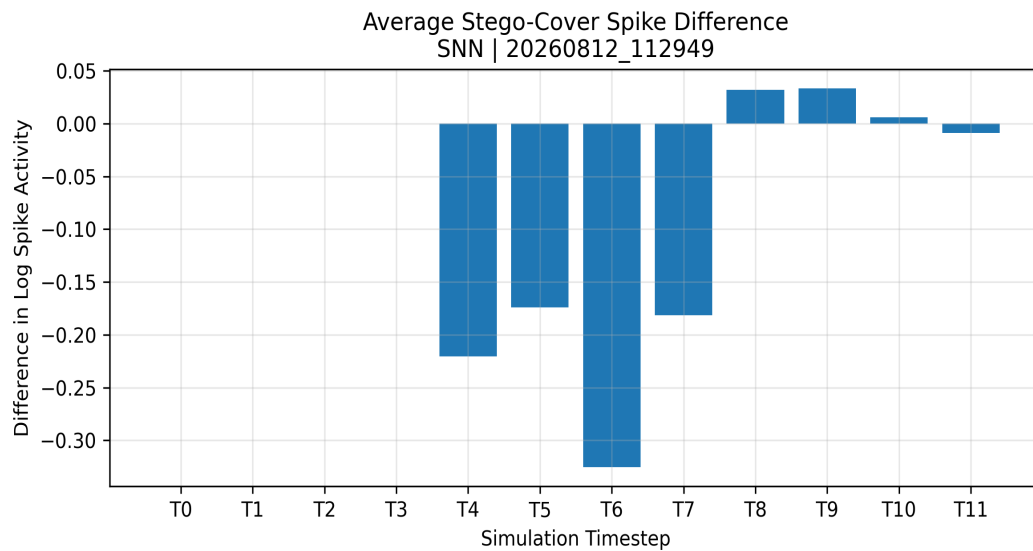
Peak Difference Timestep: T6

Maximum Absolute Difference: -0.3256

Largest Positive Shift: 0.0333

Largest Negative Shift: -0.3256

## Spike Difference Map



## Spike Activity Analysis

*The greatest spike-activity difference was observed at T6, where an absolute difference of -0.3256 was recorded. Spike activity was concentrated within the later simulation timesteps, suggesting that temporal firing behaviour diverged most strongly after membrane potentials had accumulated. The spike difference map highlights timesteps that contributed most strongly to classification differences between cover and stego images.*

## Performance Analysis

---

Mean Latency: 258.74 ms  
Median Latency: 237.83 ms  
Minimum Latency: 187.68 ms  
Maximum Latency: 2438.23 ms  
Images Processed: 700  
Total Inference Time: 181.12 s  
Throughput: 3.86 images/s

## Detection Sensitivity Analysis

---

### Payload Size Analysis

| Category | Accuracy | Precision | Recall | F1     | TNR   | FPR   |
|----------|----------|-----------|--------|--------|-------|-------|
| Small    | 71.50%   | 100.00%   | 71.50% | 83.38% | 0.00% | 0.00% |
| Medium   | 83.00%   | 100.00%   | 83.00% | 90.71% | 0.00% | 0.00% |
| Large    | 99.00%   | 100.00%   | 99.00% | 99.50% | 0.00% | 0.00% |

### Payload Type Analysis

| Category  | Accuracy | Precision | Recall | F1     | TNR   | FPR   |
|-----------|----------|-----------|--------|--------|-------|-------|
| Plain     | 84.67%   | 100.00%   | 84.67% | 91.70% | 0.00% | 0.00% |
| Encrypted | 84.33%   | 100.00%   | 84.33% | 91.50% | 0.00% | 0.00% |

### Payload Category Analysis

| Category         | Accuracy | Precision | Recall  | F1      | TNR   | FPR   |
|------------------|----------|-----------|---------|---------|-------|-------|
| Encrypted Large  | 98.00%   | 100.00%   | 98.00%  | 98.99%  | 0.00% | 0.00% |
| Encrypted Medium | 83.00%   | 100.00%   | 83.00%  | 90.71%  | 0.00% | 0.00% |
| Encrypted Small  | 72.00%   | 100.00%   | 72.00%  | 83.72%  | 0.00% | 0.00% |
| Plain Large      | 100.00%  | 100.00%   | 100.00% | 100.00% | 0.00% | 0.00% |
| Plain Medium     | 83.00%   | 100.00%   | 83.00%  | 90.71%  | 0.00% | 0.00% |
| Plain Small      | 71.00%   | 100.00%   | 71.00%  | 83.04%  | 0.00% | 0.00% |

## Research Interpretation

### *Payload Size*

Large payloads achieved the strongest performance (F1: 99.50%), while Small payloads produced the weakest results (F1: 83.38%). This suggests that detection performance improves as payload volume increases.

### *Payload Type*

Payload encryption had negligible impact on performance, with plain and encrypted payloads achieving comparable F1 scores (91.70% and 91.50%, respectively).

## **Research Insight**

The observed performance trend suggests that ImageSNN is responding primarily to embedding artefacts rather than payload content. This behaviour is consistent with findings obtained from ImageCNN despite the substantial architectural differences between the two models.

*Note: AUC and MSE metrics are not reported for sensitivity-analysis subsets because these subsets contain only stego samples. Both metrics require a mixture of cover and stego images to provide meaningful comparative measurements. Consequently, the sensitivity analysis focuses on accuracy, precision, recall and F1 score, which remain valid indicators of detection performance within individual payload groups.*

## Limitations

---

This report provides an automated assessment based on the configured steganalysis model and evaluation parameters. Detection results represent probabilistic indicators rather than definitive proof of steganographic activity. False positives and false negatives may occur, particularly when processing previously unseen datasets, alternative embedding techniques, heavily compressed images, or images that differ significantly from the training data. Results should therefore be interpreted in conjunction with additional forensic analysis, supporting evidence, and professional judgement. Model performance is influenced by dataset characteristics, payload size, image format, calibration thresholds, and other experimental factors documented elsewhere in this report.